

Property 1:

$$\mathcal{L}\{tf(t)\} = -\frac{d}{ds}F(s).$$

Property 2 (First Shifting Theorem): and using the shifting theorem

$$\mathcal{L}\{e^{ct}f(t)\} = F(s - c),$$

Property 3 (Second Shifting Theorem): Let $f(t)$ be a function of exponential type. Assume $c > 0$. Then

$$\mathcal{L}\{f(t - c)u(t - c)\} = e^{-cs}(\mathcal{L}f).$$

In the above u denotes the Heaviside function.

Property 4: (Convolution formula for Laplace transform)

$$\mathcal{L}(f * g) = \mathcal{L}(f)\mathcal{L}(g) = F(s)G(s), \quad (0.1)$$

where

$$(f * g)(t) = \int_0^t f(r)g(t - r)dr. \quad (0.2)$$

(1) (Change of Scale Property). If $\mathcal{L}\{f(t)\} = F(s)$, then show

$$\mathcal{L}\{f(at)\} = \frac{1}{a}F\left(\frac{s}{a}\right), \quad a > 0.$$

Solution: Suppose

$$\mathcal{L}\{f(t)\} = F(s) = \int_0^\infty e^{-st}f(t) dt.$$

We compute

$$\mathcal{L}\{f(at)\} = \int_0^\infty e^{-st}f(at) dt.$$

Let $u = at$, so that

$$t = \frac{u}{a}, \quad dt = \frac{du}{a}.$$

Then

$$\mathcal{L}\{f(at)\} = \int_0^\infty e^{-s\frac{u}{a}}f(u)\frac{du}{a} = \frac{1}{a} \int_0^\infty e^{-\frac{s}{a}u}f(u) du.$$

Hence

$$\boxed{\mathcal{L}\{f(at)\} = \frac{1}{a}F\left(\frac{s}{a}\right), \quad a > 0.}$$

(2) Consider the following functions:

(i) The floor function $f(t) = [t]$, where for any integer n ,

$$[t] = n \quad \text{for all } n \leq t < n + 1.$$

(ii)

$$f(t) = \frac{e^{t^2}}{e^{2t} + 1}.$$

(iii)

$$f(t) = t^2 e^{-t}.$$

Answer the following questions for each function.

- (a) Is $f(t)$ continuous on $0 \leq t < \infty$, discontinuous but piecewise continuous on $0 \leq t < \infty$, or neither?
- (b) Do there exist fixed numbers a and M such that

$$|f(t)| \leq M e^{at} \quad \text{for } 0 \leq t < \infty?$$

Solution:

(i) $f(t) = [t]$

Continuity:

The floor function is constant on each interval $[n, n+1)$ and jumps at integers. Thus it is discontinuous at integers but piecewise continuous on $[0, \infty)$.

Exponential order: We first recall the definition of exponential order.

Definition. A function f is of *exponential order* if there exist $\alpha \in \mathbb{R}$, $t_0 \in \mathbb{R}$ and $M > 0$ such that for all $t \geq t_0$,

$$|f(t)| \leq M e^{\alpha t}.$$

Equivalently, $\lim_{t \rightarrow \infty} e^{-\alpha t} |f(t)| = 0$.

As $[t] \leq t \leq e^t$ for $t > 0$ we can compute

$$\lim_{t \rightarrow \infty} e^{-2t} |[t]| \leq \lim_{t \rightarrow \infty} e^{-2t} e^t = \lim_{t \rightarrow \infty} e^{-t} = 0. \quad (0.3)$$

Thus it is of exponential order $\alpha = 2$.

(ii) $f(t) = \frac{e^{t^2}}{e^{2t} + 1}$

Continuity:

Both numerator and denominator are continuous and $e^{2t} + 1 > 0$. Hence f is continuous on $[0, \infty)$.

Exponential order: Suppose f is of exponential order. Then there exists $\alpha \in \mathbb{R}$ such that $\lim_{t \rightarrow \infty} e^{-\alpha t} |f(t)| = 0$. We now compute

$$\lim_{t \rightarrow \infty} e^{-\alpha t} |f(t)| = \lim_{t \rightarrow \infty} \frac{e^{-\alpha t} e^{t^2}}{e^{2t} + 1}$$

Using L' Hospital's rule we obtain

$$\begin{aligned} \lim_{t \rightarrow \infty} e^{-\alpha t} |f(t)| &= \lim_{t \rightarrow \infty} \frac{e^{-\alpha t} e^{t^2}}{e^{2t} + 1} = \lim_{t \rightarrow \infty} \frac{e^{-\alpha t + t^2} (-\alpha + 2t)}{2e^{2t}} \\ &= \lim_{t \rightarrow \infty} e^{-\alpha t + t^2 - 2t} (-\alpha + 2t) = \infty \end{aligned}$$

Hence we arrive at a contradiction and hence f is not of exponential order.

(iii) $f(t) = t^2 e^{-t}$

Continuity:

Product of continuous functions $\Rightarrow$ continuous.

Exponential order: We compute

$$\lim_{t \rightarrow \infty} e^{-t} |f(t)| = \lim_{t \rightarrow \infty} e^{-t} t^2 e^{-t} = \lim_{t \rightarrow \infty} \frac{t^2}{e^{2t}} = 0, \quad (0.4)$$

by using L' Hospitals rule. Hence f is of exponential order.

(3) Find the Laplace transforms of the following functions:

- (a) $t^m \cosh(bt)$
- (b) $e^t \sin(at)$
- (c) $te^t \sin(at)$
- (d) $\frac{\sin t \cosh t}{t}$
- (e) $5e^{-7t} + t + 2e^{2t}$
- (f)

$$f(t) = \begin{cases} \sin 3t, & 0 < t < \pi, \\ 0, & t > \pi. \end{cases}$$

Solution:

(a) $\mathcal{L}\{t^m \cosh(bt)\}$

From the definition of cosh function we can write

$$f(t) = t^m \cosh(bt) = \frac{t^m}{2} (e^{bt} + e^{-bt})$$

Taking Laplace transform and using linearity property of the Laplace transform we obtain

$$\mathcal{L}\{f(t)\} = \frac{1}{2} \mathcal{L}\{t^m e^{bt}\} + \frac{1}{2} \mathcal{L}\{t^m e^{-bt}\}$$

Recalling that $\mathcal{L}\{t^m\} = \frac{m!}{s^{m+1}}$ and using the first shifting theorem we can write

$$\begin{aligned} \mathcal{L}\{f(t)\} &= \frac{1}{2} \left\{ \frac{m!}{(s-b)^{m+1}} + \frac{m!}{(s+b)^{m+1}} \right\} \\ &= \frac{m!}{2} \left\{ \frac{1}{(s-b)^{m+1}} + \frac{1}{(s+b)^{m+1}} \right\}. \end{aligned}$$

(b) $\mathcal{L}\{e^t \sin(at)\}$

Let $f(t) = \sin(at)$. Then we know that

$$\mathcal{L}\{f(t)\} = \mathcal{L}\{\sin(at)\} = \frac{a}{s^2 + a^2} = F(s), \text{ (say).}$$

Now, using first shifting theorem we obtain

$$\mathcal{L}\{e^t \sin(at)\} = \mathcal{L}\{e^t f(t)\} = F(s-1) = \frac{a}{(s-1)^2 + a^2}. \quad (0.5)$$

(c) $\mathcal{L}\{te^t \sin(at)\}$

Let $f(t) = e^t \sin(at)$. Then from equation (0.5) we can write

$$\mathcal{L}\{e^t \sin(at)\} = \frac{a}{(s-1)^2 + a^2}. \quad (0.6)$$

Now, using Property 1 described above we obtain

$$\mathcal{L}\{tf(t)\} = -\frac{d}{ds}F(s),$$

Finally we obtain

$$\boxed{\mathcal{L}\{te^t \sin(at)\} = -\frac{d}{ds} \left(\frac{a}{(s-1)^2 + a^2} \right)}.$$

(d) $\mathcal{L}\left\{\frac{\sin t \cosh t}{t}\right\}$

In order to find the Laplace transform of such function, we use the result:

$$\mathcal{L}\left\{\frac{F(t)}{t}\right\}(s) = \int_s^\infty \mathcal{L}\{F(t)\}(u) du.$$

Let $F(t) = \sin t \cosh t$. Then we have

$$\mathcal{L}\left\{\frac{\sin t \cosh t}{t}\right\}(s) = \int_s^\infty \mathcal{L}\{\sin t \cosh t\}(u) du.$$

Therefore, we first compute $\mathcal{L}\{\sin t \cosh t\}$.

Note that

$$\cosh t = \frac{e^t + e^{-t}}{2}.$$

Thus

$$\sin t \cosh t = \frac{1}{2}(e^t \sin t + e^{-t} \sin t).$$

Recalling that $\mathcal{L}\{\sin t\} = \frac{1}{s^2+1}$, using first shifting theorem we obtain

$$\begin{aligned} \mathcal{L}\{\sin t \cosh t\} &= \frac{1}{2} (\mathcal{L}\{e^t \sin t\} + \mathcal{L}\{e^{-t} \sin t\}) \\ &= \frac{1}{2} \left(\frac{1}{(s-1)^2 + 1} + \frac{1}{(s+1)^2 + 1} \right) \end{aligned}$$

Therefore, we obtain

$$\begin{aligned} \mathcal{L}\left\{\frac{\sin t \cosh t}{t}\right\}(s) &= \int_s^\infty \mathcal{L}\{\sin t \cosh t\}(u) du \\ &= \frac{1}{2} \int_s^\infty \left(\frac{1}{(u-1)^2 + 1} + \frac{1}{(u+1)^2 + 1} \right) du \\ &= \frac{1}{2} \left[\tan^{-1} \left(\frac{1}{u-1} \right) + \tan^{-1} \left(\frac{1}{u+1} \right) \right]_s^\infty \\ &= \frac{1}{2} \left[\tan^{-1} \left(\frac{1}{s-1} \right) + \tan^{-1} \left(\frac{1}{s+1} \right) \right]. \end{aligned}$$

(e) $\mathcal{L}\{5e^{-7t} + t + 2e^{2t}\}$

Using linearity property of Laplace transform we obtain

$$\mathcal{L}\{5e^{-7t} + t + 2e^{2t}\} = 5\mathcal{L}\{e^{-7t}\} + \mathcal{L}\{t\} + 2\mathcal{L}\{e^{2t}\} = \boxed{\frac{5}{s+7} + \frac{1}{s^2} + \frac{2}{s-2}}.$$

(f)

$$f(t) = \begin{cases} \sin(3t), & 0 < t < \pi, \\ 0, & t > \pi. \end{cases}$$

$$\begin{aligned} (\mathcal{L}f)(s) &= \int_0^{\infty} e^{-st} f(t) dt \\ &= \int_0^{\pi} e^{-st} \sin 3t dt \\ &= -\frac{1}{s} e^{-st} \sin 3t \Big|_0^{\pi} + \frac{3}{s} \int_0^{\pi} e^{-st} \cos 3t dt \\ &= 0 + \frac{3}{s} \left\{ -\frac{1}{s} e^{-st} \cos 3t \Big|_0^{\pi} - \frac{3}{s} \int_0^{\pi} e^{-st} \sin 3t dt \right\} \\ &= \frac{3}{s^2} \left\{ -e^{-\pi s}(-1) + 1 - \frac{9}{s^2} (\mathcal{L}f)(s) \right\}. \end{aligned}$$

$$\begin{aligned} \Rightarrow \left(1 + \frac{9}{s^2}\right) (\mathcal{L}f)(s) &= \frac{3}{s^2} \{1 + e^{-\pi s}\} \\ \Rightarrow (\mathcal{L}f)(s) &= \frac{3}{s^2 + 9} \{1 + e^{-\pi s}\}. \end{aligned}$$

(4) Find the inverse Laplace transforms of the following functions:

- (a) $\tan^{-1}\left(\frac{a}{s}\right)$
- (b) $\log\left(\frac{s^2+1}{(s+1)^2}\right)$
- (c) $\frac{1-3s}{s^2+8s+21}$
- (d) $\frac{s+2}{(s^2+4s-5)^2}$
- (e) $\frac{se^{-\pi s}}{s^2+4}$
- (f) $\frac{(1-e^{-2s})(1-3e^{-2s})}{s^2}$

Solution:

(a)

Suppose $F(s) = \tan^{-1}\left(\frac{a}{s}\right)$. We use the property

$$\mathcal{L}\{tf(t)\} = -\frac{d}{ds}F(s).$$

Then

$$-\frac{dF}{ds} = -\frac{1}{1 + \frac{a^2}{s^2}} \cdot \frac{a}{s^2}.$$

Then

$$\frac{dF}{ds} = \frac{a}{s^2 + a^2} = \mathcal{L}\{\sin(at)\}.$$

Hence,

$$\mathcal{L}\{tf(t)\} = \mathcal{L}\{\sin(at)\}.$$

Taking inverse Laplace transform,

$$tf(t) = \sin(at),$$

so that

$$f(t) = \frac{\sin(at)}{t}.$$

(b)

Suppose

$$F(s) = \log\left(\frac{s^2 + 1}{(s + 1)^2}\right).$$

Again, we use the property

$$\mathcal{L}\{tf(t)\} = -F'(s).$$

Differentiating with respect to s we obtain

$$\frac{dF}{ds} = \frac{d}{ds} [\log(s^2 + 1) - 2\log(s + 1)].$$

$$= \frac{2s}{s^2 + 1} - \frac{2}{s + 1}.$$

Thus,

$$\frac{dF}{ds} = 2\left(\frac{s}{s^2 + 1} - \frac{1}{s + 1}\right).$$

Now,

$$\frac{s}{s^2 + 1} = \mathcal{L}\{\cos t\}, \quad \frac{1}{s + 1} = \mathcal{L}\{e^{-t}\}.$$

Therefore,

$$\frac{dF}{ds} = \mathcal{L}\{2\cos t - 2e^{-t}\}.$$

Using the property,

$$\mathcal{L}\{tf(t)\} = -F'(s) = \mathcal{L}\{2e^{-t} - 2\cos t\}.$$

Taking inverse Laplace transform,

$$tf(t) = 2e^{-t} - 2\cos t.$$

Hence,

$$f(t) = \frac{2}{t} (e^{-t} - \cos t).$$

(c) We compute the inverse Laplace transform:

$$\mathcal{L}^{-1} \left\{ \frac{1 - 3s}{s^2 + 8s + 21} \right\}.$$

We can write

$$F(s) = \frac{13 - 3(s + 4)}{(s + 4)^2 + 5}.$$

We can decompose F as

$$F(s) = \frac{13}{(s + 4)^2 + 5} - \frac{3(s + 4)}{(s + 4)^2 + 5}.$$

Recall the standard Laplace transform pairs:

1. $\mathcal{L}\{e^{-4t} \cos(\sqrt{5}t)\} = \frac{s + 4}{(s + 4)^2 + 5}.$
2. $\mathcal{L}\{e^{-4t} \sin(\sqrt{5}t)\} = \frac{\sqrt{5}}{(s + 4)^2 + 5}.$

For the first term, we compare:

$$\frac{13}{(s + 4)^2 + 5} = 13 \cdot \frac{1}{(s + 4)^2 + 5} = 13 \cdot \frac{1}{\sqrt{5}} \cdot \frac{\sqrt{5}}{(s + 4)^2 + 5}.$$

Thus, the inverse Laplace transform is:

$$\mathcal{L}^{-1} \left\{ \frac{13}{(s + 4)^2 + 5} \right\} = \frac{13}{\sqrt{5}} e^{-4t} \sin(\sqrt{5}t).$$

For the second term, we use the cosine transform directly:

$$\mathcal{L}^{-1} \left\{ \frac{3(s + 4)}{(s + 4)^2 + 5} \right\} = 3e^{-4t} \cos(\sqrt{5}t).$$

Combine the inverse transforms, taking care of the negative sign for the second term:

$$f(t) = \frac{13}{\sqrt{5}} e^{-4t} \sin(\sqrt{5}t) - 3e^{-4t} \cos(\sqrt{5}t).$$

Final Answer

$$\boxed{f(t) = e^{-4t} \left(\frac{13}{\sqrt{5}} \sin(\sqrt{5}t) - 3 \cos(\sqrt{5}t) \right)}$$

(d) We compute

$$\mathcal{L}^{-1} \left\{ \frac{s+2}{(s^2+4s-5)^2} \right\}.$$

Given

$$F(s) = \frac{s+2}{(s^2+4s-5)^2}.$$

Note that

$$s^2+4s-5 = (s+2)^2 - 9,$$

so

$$F(s) = \frac{s+2}{((s+2)^2-9)^2}.$$

Observe that

$$\frac{d}{ds} \left(\frac{1}{(s+2)^2-9} \right) = -\frac{2(s+2)}{((s+2)^2-9)^2}.$$

Hence

$$F(s) = -\frac{1}{2} \frac{d}{ds} \left(\frac{1}{(s+2)^2-9} \right).$$

Since

$$\mathcal{L} \left\{ \frac{e^{-2t} \sinh(3t)}{3} \right\} = \frac{1}{(s+2)^2-9},$$

we have

$$F(s) = -\frac{1}{2} \frac{d}{ds} \left[\mathcal{L} \left\{ \frac{e^{-2t} \sinh(3t)}{3} \right\} \right].$$

Using the identity

$$\mathcal{L}\{tf(t)\} = -\frac{d}{ds} \mathcal{L}\{f(t)\},$$

we get

$$F(s) = \frac{1}{6} \mathcal{L}\{te^{-2t} \sinh(3t)\}.$$

Therefore,

$$\mathcal{L}^{-1} \left\{ \frac{s+2}{(s^2+4s-5)^2} \right\} = \frac{te^{-2t} \sinh(3t)}{6}.$$

(e)

$$\mathcal{L}^{-1} \left(\frac{s}{s^2+4} \right) = \cos(2t).$$

Using Property 3, we have

$$\boxed{\mathcal{L}^{-1} \left(\frac{se^{-\pi s}}{s^2+4} \right) = u(t-\pi) \cos(2(t-\pi)).}$$

(f)

Expand:

$$\frac{(1 - e^{-2s})(1 - 3e^{-2s})}{s^2} = \frac{1}{s^2} - \frac{4e^{-2s}}{s^2} + \frac{3e^{-4s}}{s^2}.$$

Since

$$\mathcal{L}^{-1}\left\{\frac{1}{s^2}\right\} = t,$$

using Property 3 we obtain

$$\boxed{f(t) = t - 4u(t-2)(t-2) + 3u(t-4)(t-4).}$$

(5) Use Laplace transforms to solve the following initial value problems

(a)

$$y'' + 4y = \cos 2t, \quad y(0) = 0, \quad y'(0) = 1.$$

(b)

$$y'' + 4y' = \cos(t-3) + 4t, \quad y(3) = 0, \quad y'(3) = 7.$$

(c)

$$y'' + 3y' + 2y = f(t), \quad y(0) = 0, \quad y'(0) = 0,$$

where

$$f(t) = \begin{cases} 4t, & 0 < t < 1, \\ 8, & t > 1. \end{cases}$$

(d)

$$y'' + 4y = f(t), \quad y(0) = 4, \quad y'(0) = 4,$$

where

$$f(t) = \begin{cases} 8 \sin t, & 0 < t < \pi, \\ 0, & t > \pi. \end{cases}$$

Solution:

$$(a) \quad y'' + 4y = \cos 2t, \quad y(0) = 0, \quad y'(0) = 1$$

Taking Laplace transform:

$$\mathcal{L}\{y''\} = s^2Y(s) - sy(0) - y'(0)$$

Using the initial conditions we obtain

$$\mathcal{L}\{y''\} = s^2Y(s) - sy(0) - y'(0) = s^2Y(s) - 1.$$

Applying Laplace transform on both sides of the given differential equation we get

$$\mathcal{L}\{y''\} + 4\mathcal{L}\{y\} = \mathcal{L}\{\cos 2t\} \tag{0.7}$$

Thus

$$\begin{aligned}
s^2 Y(s) - 1 + 4Y(s) &= \frac{s}{s^2 + 4}. \\
(s^2 + 4)Y(s) &= \frac{s}{s^2 + 4} + 1. \\
Y(s) &= \frac{s}{(s^2 + 4)^2} + \frac{1}{s^2 + 4}.
\end{aligned} \tag{0.8}$$

Now taking inverse Laplace transform we obtain

$$\begin{aligned}
\mathcal{L}^{-1}\left\{\frac{1}{s^2 + 4}\right\} &= \frac{1}{2} \sin 2t, \\
\mathcal{L}^{-1}\left\{\frac{s}{(s^2 + 4)^2}\right\} &= \mathcal{L}^{-1}\left\{\frac{s}{(s^2 + 4)} \frac{1}{s^2 + 4}\right\}
\end{aligned}$$

Since

$$\mathcal{L}^{-1}\left\{\frac{s}{s^2 + 4}\right\} = \cos 2t, \tag{0.9}$$

by using Laplace transform for convolutions we obtain

$$\mathcal{L}^{-1}\left\{\frac{s}{(s^2 + 4)^2}\right\} = \frac{1}{2} \sin 2t * \cos 2t$$

We now compute

$$(\cos 2t * \sin 2t)(t) = \int_0^t \cos(2(t - \tau)) \sin(2\tau) d\tau.$$

Using the identity

$$\cos A \sin B = \frac{1}{2} [\sin(A + B) - \sin(A - B)],$$

with $A = 2(t - \tau)$ and $B = 2\tau$, we obtain

$$\cos(2(t - \tau)) \sin(2\tau) = \frac{1}{2} [\sin(2t) - \sin(2t - 4\tau)].$$

Therefore,

$$(\cos 2t * \sin 2t)(t) = \frac{1}{2} \int_0^t [\sin(2t) - \sin(2t - 4\tau)] d\tau.$$

Now compute the integrals:

$$\int_0^t \sin(2t) d\tau = t \sin(2t),$$

and

$$\int_0^t \sin(2t - 4\tau) d\tau = \frac{1}{4} (\cos(2t - 4t) - \cos(2t)) = \frac{1}{4} (\cos(-2t) - \cos(2t)) = 0.$$

Hence,

$$(\cos 2t * \sin 2t)(t) = \frac{1}{2} t \sin(2t).$$

$$\boxed{(\cos 2t * \sin 2t)(t) = \frac{t}{2} \sin(2t)}$$

Therefore, applying inverse Laplace transform in equation (0.8) we obtain

$$y(t) = \mathcal{L}^{-1}\{Y(s)\} = \mathcal{L}^{-1}\left\{\frac{s}{(s^2 + 4)^2}\right\} + \mathcal{L}^{-1}\left\{\frac{s}{(s^2 + 4)}\right\} \quad (0.10)$$

$$\boxed{y(t) = \frac{t}{4} \sin 2t + \frac{1}{2} \sin 2t.}$$

(b) $y'' + 4y' = \cos(t - 3) + 4t, \quad y(3) = 0, \quad y'(3) = 7$

Let $x = t - 3$. Define $u(x) = y(t)$.

Then

$$u''(x) + 4u'(x) = \cos x + 4(x + 3).$$

Initial conditions:

$$u(0) = y(0 + 3) = 0, \quad u'(0) = y'(0 + 3) = 7.$$

Taking Laplace transform:

$$\begin{aligned} s^2 U(s) - su(0) - u'(0) + 4(sU(s) - u(0)) &= \frac{s}{s^2 + 1} + \frac{4}{s^2} + \frac{12}{s} \\ \Rightarrow (s^2 + 4s)U(s) - 7 &= \frac{s}{s^2 + 1} + \frac{4}{s^2} + \frac{12}{s} \\ \Rightarrow U(s) &= \frac{7}{s(s + 4)} + \frac{s}{(s^2 + 1)s(s + 4)} + \frac{4}{s^2 s(s + 4)} + \frac{12}{ss(s + 4)} \\ \Rightarrow U(s) &= \frac{1}{(s^2 + 1)(s + 4)} + \frac{4 + 12s + 7s^2}{s^3(s + 4)}. \end{aligned}$$

Partial fractioning each of the terms in our transform gives us the following:

$$\begin{aligned} \frac{1}{(s + 4)(s^2 + 1)} &= \frac{1}{17} \frac{1}{s + 4} + \frac{1}{17} \left(\frac{-s + 4}{s^2 + 1} \right) \\ \frac{4 + 12s + 7s^2}{s^3(s + 4)} &= \frac{1}{s^3} + \frac{11}{4} \frac{1}{s^2} + \frac{17}{16} \frac{1}{s} - \frac{17}{16} \frac{1}{s + 4} \end{aligned}$$

This can be seen as follows:

$$\begin{aligned} \frac{1}{(s + 4)(s^2 + 1)} &= \frac{A}{s + 4} + \left(\frac{Bs + C}{s^2 + 1} \right) \\ \Rightarrow 1 &= A(s^2 + 1) + (Bs + C)(s + 4) \end{aligned} \quad (0.11)$$

Putting $s = -4$ we obtain $A = \frac{1}{17}$. Comparing the coefficient of s^2 we obtain $A + B = 0$. Consequently, $B = -\frac{1}{17}$.

Differentiating both sides with respect to s we obtain

$$0 = 2sA + Bs + C + B(s + 4). \quad (0.12)$$

After putting $s = 0$ we obtain

$$0 = C - \frac{4}{17}, \quad (0.13)$$

and hence $C = \frac{4}{17}$.

Similarly we can express

$$\frac{4 + 12s + 7s^2}{s^3(s + 4)} = \frac{A}{s} + \frac{B}{s^2} + \frac{C}{s^3} + \frac{D}{s + 4}, \quad (0.14)$$

and we can find the values of A, B, C and D .

Plugging these into our transform and combining like terms gives us

$$\begin{aligned} U(s) &= \frac{1}{s^3} + \frac{11}{4} \frac{1}{s^2} + \frac{17}{16} \frac{1}{s} - \frac{273}{272} \frac{1}{s + 4} + \frac{1}{17} \left(\frac{-s + 4}{s^2 + 1} \right) \\ &= \frac{1 \cdot 2!}{2!} \frac{1}{s^3} + \frac{11}{4} \frac{1}{s^2} + \frac{17}{16} \frac{1}{s} - \frac{273}{272} \frac{1}{s + 4} + \frac{1}{17} \left(\frac{-s}{s^2 + 1} + \frac{4}{s^2 + 1} \right) \end{aligned}$$

Now, taking the inverse transform will give the solution to our new IVP.

$$u(x) = \frac{1}{2}x^2 + \frac{11}{4}x + \frac{17}{16} - \frac{273}{272}e^{-4x} + \frac{1}{17}(4\sin(x) - \cos(x))$$

$$y(t) = y(x + 3) = u(x) = u(t - 3).$$

So, the solution to the original IVP is,

$$y(t) = \frac{1}{2}(t - 3)^2 + \frac{11}{4}(t - 3) + \frac{17}{16} - \frac{273}{272}e^{-4(t-3)} + \frac{1}{17}(4\sin(t - 3) - \cos(t - 3))$$

$$y(t) = \frac{1}{2}t^2 - \frac{1}{4}t - \frac{43}{16} - \frac{273}{272}e^{-4(t-3)} + \frac{1}{17}(4\sin(t - 3) - \cos(t - 3))$$

$$(c) \ y'' + 3y' + 2y = f(t), \quad y(0) = 0, \ y'(0) = 0$$

$$f(t) = \begin{cases} 4t, & 0 < t < 1, \\ 8, & t > 1. \end{cases}$$

Write using Heaviside function:

$$f(t) = 4t + (8 - 4t)u(t - 1).$$

Taking Laplace transform of the differential equation:

$$\mathcal{L}\{y''\} + 3\mathcal{L}\{y'\} + 2\mathcal{L}\{y\} = \mathcal{L}\{f(t)\}.$$

Using

$$\mathcal{L}\{y''\} = s^2Y(s) - sy(0) - y'(0),$$

$$\mathcal{L}\{y'\} = sY(s) - y(0),$$

and $y(0) = 0$, $y'(0) = 0$, we get

$$(s^2 + 3s + 2)Y(s) = \mathcal{L}\{f(t)\} \Rightarrow Y(s) = \frac{\mathcal{L}\{f(t)\}}{(s^2 + 3s + 2)}$$

We now compute $\mathcal{L}\{f(t)\}$. First,

$$\mathcal{L}\{4t\} = \frac{4}{s^2}.$$

Next use Property 3:

$$\mathcal{L}\{u(t-a)g(t-a)\} = e^{-as}G(s).$$

We compute

$$\mathcal{L}\{[4 - 4(t-1)]u(t-1)\} = e^{-s}\mathcal{L}\{4 - 4t\}.$$

Now,

$$\mathcal{L}\{4\} = \frac{4}{s}, \quad \mathcal{L}\{4t\} = \frac{4}{s^2}.$$

Thus,

$$\mathcal{L}\{4 - 4t\} = \frac{4}{s} - \frac{4}{s^2}.$$

Therefore,

$$\mathcal{L}\{f(t)\} = \frac{4}{s^2} + e^{-s} \left(\frac{4}{s} - \frac{4}{s^2} \right).$$

Hence,

$$\begin{aligned} Y(s) &= \frac{1}{(s+1)(s+2)} \left[\frac{4}{s^2} + e^{-s} \left(\frac{4}{s} - \frac{4}{s^2} \right) \right] \\ &= \frac{4}{s^2(s+1)(s+2)} + e^{-s} \frac{\frac{4}{s} - \frac{4}{s^2}}{(s+1)(s+2)}. \end{aligned}$$

First term:

$$\frac{4}{s^2(s+1)(s+2)}$$

Use partial fractions:

$$\frac{4}{s^2(s+1)(s+2)} = -\frac{6}{s} + \frac{2}{s^2} + \frac{4}{s+1} + \frac{2}{s+2}.$$

Taking inverse Laplace:

$$\mathcal{L}^{-1}\left(\frac{4}{s^2(s+1)(s+2)}\right) = -6 + 2t + 4e^{-t} + 2e^{-2t}.$$

Second Term:

$$e^{-s} \frac{\frac{4}{s} - \frac{4}{s^2}}{(s+1)(s+2)}.$$

Let

$$H(s) = \frac{\frac{4}{s} - \frac{4}{s^2}}{(s+1)(s+2)}.$$

After partial fractions and simplification,

$$\mathcal{L}^{-1}\{H(s)\} = 2 - 2t - 4e^{-t} + 2e^{-2t}.$$

By shifting theorem,

$$\mathcal{L}^{-1}\{e^{-s}H(s)\} = u(t-1)[2 - 2(t-1) - 4e^{-(t-1)} + 2e^{-2(t-1)}].$$

$$y(t) = -6 + 2t + 4e^{-t} + 2e^{-2t} \\ + u(t-1)[2 - 2(t-1) - 4e^{-(t-1)} + 2e^{-2(t-1)}].$$

$$(d) \ y'' + 4y = f(t), \quad y(0) = 4, \ y'(0) = 4$$

$$f(t) = \begin{cases} 8 \sin t, & 0 < t < \pi, \\ 0, & t > \pi. \end{cases}$$

We solve the initial value problem

$$y'' + 4y = f(t), \quad y(0) = 4, \quad y'(0) = 4,$$

where

$$f(t) = \begin{cases} 8 \sin t, & 0 < t < \pi, \\ 0, & t > \pi. \end{cases}$$

Since the forcing term vanishes after $t = \pi$, we write

$$f(t) = 8 \sin t [1 - u(t - \pi)].$$

Thus

$$f(t) = 8 \sin t - 8 \sin t u(t - \pi).$$

Note that

$$\sin t = \sin((t - \pi) + \pi) = -\sin(t - \pi).$$

Hence

$$\sin t u(t - \pi) = -\sin(t - \pi) u(t - \pi).$$

Therefore,

$$f(t) = 8 \sin t + 8 \sin(t - \pi) u(t - \pi).$$

Taking Laplace transform of the differential equation:

$$\mathcal{L}\{y''\} + 4\mathcal{L}\{y\} = \mathcal{L}\{f(t)\}.$$

Using

$$\mathcal{L}\{y''\} = s^2 Y(s) - sy(0) - y'(0),$$

and the given initial conditions,

$$(s^2 Y - 4s - 4) + 4Y = \mathcal{L}\{f(t)\}.$$

Thus,

$$(s^2 + 4)Y(s) - 4s - 4 = \mathcal{L}\{f(t)\}.$$

Hence

$$Y(s) = \frac{\mathcal{L}\{f(t)\} + 4s + 4}{s^2 + 4}.$$

First,

$$\mathcal{L}\{8 \sin t\} = \frac{8}{s^2 + 1}.$$

Next, using Property 3:

$$\mathcal{L}\{u(t - a)g(t - a)\} = e^{-as}G(s),$$

we obtain

$$\mathcal{L}\{8 \sin(t - \pi)u(t - \pi)\} = 8e^{-\pi s} \frac{1}{s^2 + 1}.$$

Thus,

$$\mathcal{L}\{f(t)\} = \frac{8}{s^2 + 1} + \frac{8e^{-\pi s}}{s^2 + 1}.$$

$$Y(s) = \frac{8}{(s^2 + 1)(s^2 + 4)} + \frac{8e^{-\pi s}}{(s^2 + 1)(s^2 + 4)} + \frac{4s + 4}{s^2 + 4}.$$

First Term: We decompose

$$\frac{8}{(s^2 + 1)(s^2 + 4)} = \frac{8}{3} \frac{1}{s^2 + 1} - \frac{8}{3} \frac{1}{s^2 + 4}.$$

Hence,

$$\mathcal{L}^{-1} \left(\frac{8}{(s^2 + 1)(s^2 + 4)} \right) = \frac{8}{3} \sin t - \frac{4}{3} \sin(2t).$$

Second Term:

$$e^{-\pi s} \frac{8}{(s^2 + 1)(s^2 + 4)}.$$

Thus, using Property 3 we obtain

$$\mathcal{L}^{-1} \left(e^{-\pi s} \frac{8}{(s^2 + 1)(s^2 + 4)} \right) = u(t - \pi) \left[\frac{8}{3} \sin(t - \pi) - \frac{4}{3} \sin(2(t - \pi)) \right].$$

Using

$$\sin(t - \pi) = -\sin t, \quad \sin(2(t - \pi)) = \sin(2t),$$

we obtain

$$u(t - \pi) \left[-\frac{8}{3} \sin t - \frac{4}{3} \sin(2t) \right].$$

Third Term:

$$\frac{4s + 4}{s^2 + 4} = 4 \frac{s}{s^2 + 4} + 4 \frac{1}{s^2 + 4}.$$

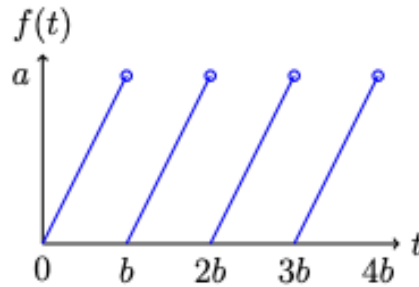
Thus

$$\mathcal{L}^{-1} \left(\frac{4s + 4}{s^2 + 4} \right) = 4 \cos(2t) + 2 \sin(2t).$$

Combining all parts,

$$\boxed{y(t) = 4 \cos(2t) + 2 \sin(2t) + \frac{8}{3} \sin t - \frac{4}{3} \sin(2t) + u(t - \pi) \left[-\frac{8}{3} \sin t - \frac{4}{3} \sin(2t) \right].}$$

(6) Laplace Transform of the Sawtooth Curve



Solution: The sawtooth function increases linearly from 0 to a on each interval $[0, b)$ and repeats with period b .

Thus

$$f(t) = \frac{a}{b}t, \quad 0 < t < b,$$

and periodic of period b .

Theorem 0.1. Suppose that $f : [0, \infty) \rightarrow \mathbb{R}$ is a periodic function of period $T > 0$, i.e.

$$f(t + T) = f(t) \quad \text{for all } t \geq 0.$$

If the Laplace transform of f exists, then

$$F(s) = \frac{\int_0^T f(t)e^{-st} dt}{1 - e^{-sT}}. \quad (0.15)$$

Proof. We have

$$\begin{aligned} F(s) &= \int_0^\infty f(t)e^{-st} dt \\ &= \sum_{n=0}^\infty \int_{nT}^{(n+1)T} f(t)e^{-st} dt \\ &= \sum_{n=0}^\infty \int_0^T f(u + nT)e^{-s(u+nT)} du, \quad \text{where } u = t - nT \\ &= \sum_{n=0}^\infty e^{-snT} \int_0^T f(u)e^{-su} du \\ &= \left(\int_0^T f(u)e^{-su} du \right) \sum_{n=0}^\infty e^{-snT} \\ &= \frac{\int_0^T f(u)e^{-su} du}{1 - e^{-sT}}. \end{aligned}$$

The last line follows from the fact that

$$\sum_{n=0}^\infty e^{-snT}$$

is a geometric series with common ratio $e^{-sT} < 1$ for $s > 0$. □

Compute:

$$\int_0^b e^{-st} \frac{a}{b} t dt = \frac{a}{b} \left[\frac{1}{s^2} - \frac{be^{-bs}}{s} - \frac{e^{-bs}}{s^2} \right].$$

Therefore

$$\boxed{\mathcal{L}\{f(t)\} = \frac{a}{b} \frac{\frac{1}{s^2} - \frac{be^{-bs}}{s} - \frac{e^{-bs}}{s^2}}{1 - e^{-bs}}}.$$